

Vitamins

vitamins

vital (vita)

Chemical Rxn

amine

NH-group

→ Organic Compounds

→ Micro Nutrients

→ Energy

→ Vital chemical control

- 1881 - 82

लूनीन → vitamin

→ हूफकिंस gives विटामिन थैरी
(vitamin theory)

→ 1912 → कैशमियर फंक

↓
vitamin discovery

Vitamin

Fat soluble

→ K
→ E
→ D
→ A

Water Soluble

→ B
→ C

Organ

→ Liver → store room of vitamin

↳ fat soluble

(K, E, D, A)

→ ~~urine~~ urine, sweat → B, C

vitamin . A :- Hopkins

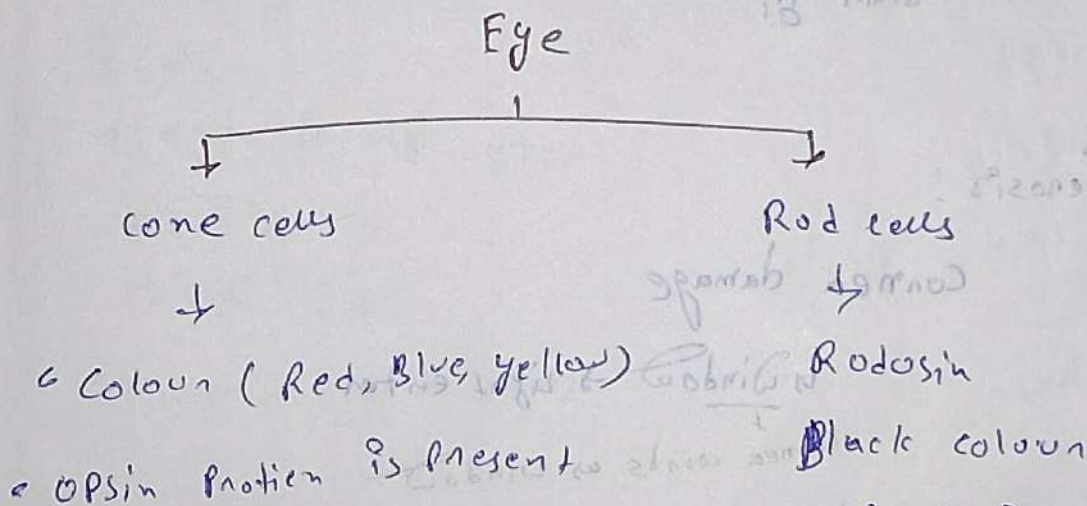
→ chemical name → Retinol . → पीला कलर

→ helps in making WBC

↓
Pale yellowish liquid

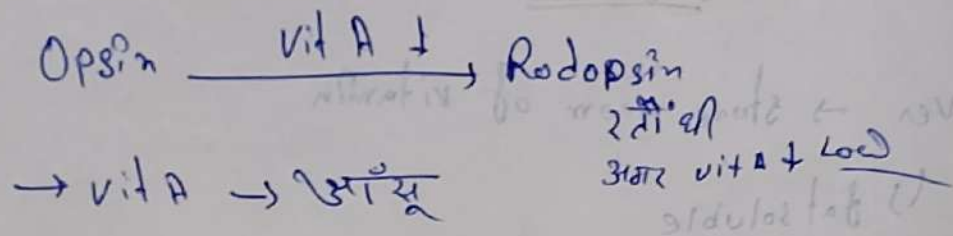
→ Immunity Boost

→ Eye vision



अंधों में दूरने में मदद करता है

Opsin + Vitamin A → Rodopsin



Disease :-

Xenophthalmia → आँसू (x)
 Dry (आँसू)

बच्ची → Blindness

माँ का दूध -) Vitamin A

Night blindness

→ Nyctalopia

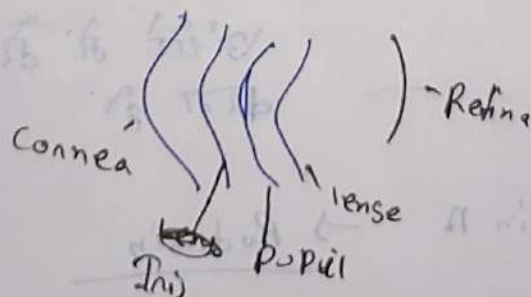
→ सूरज ढलने के बाद ठिगना बंद हो जाता है

Corneal Xerosis !

Cornea damage

↳ Window → Light enter

Cornea works as window



Sources of vitamin

Plant

हरी / पीली

- Green (Spinach) (mustard Saag)

- बैंगी, पालक, सरसो का साग

- पपीता, आम, melon, Banana

→ β -carotene Liver $\rightarrow$ vit A

Animal

Fish Liver oil

Egg

Milk Product

Vitamin - D

- calciferol

→ Hopkins & Edward Mellanbee

→ Skin

→ धूप में नहीं होता

→ Active

→ $Ca^{2+} \downarrow \uparrow$

→ Hormonal vit (PTH)

Skin में Ergosterol $\rightarrow$ UV rays पड़ने पर $\rightarrow$ D₃ में change

Body में vit D dead पड़ता होता है

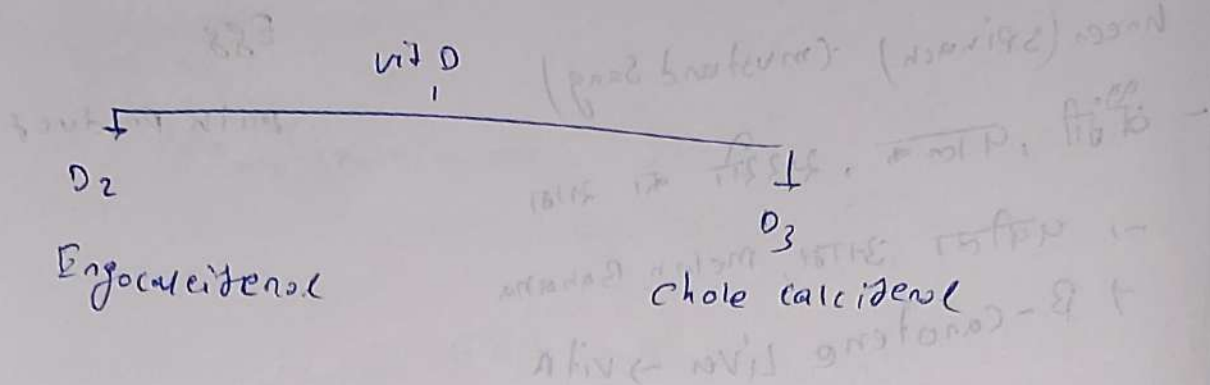
जब तक छुप न पड़े

vit D

→ co-factor

→ selenium

110 units/day



Disease

• Rickets :-

↓

Specific in child

↓

Late teeth → skull bones weak

• Bow legs

→ Adults

→ femur → Bend

• Knock knee

• Osteomalacia :

→ Adult

→ Density

→ Bend, Weak

→ Back bone

Osteoporosis :- Pores in Bone

Sources

Plant

→ Nature

→ sunlight

→ ~~माला~~

→ ~~कोले~~

→ ~~मशरूम~~

→ ~~किच~~

Animal

Eggs, chicken, meat

milk product

Vitamin - E

- Tocopherol

→ 2nd → Evans & Bishop

→ Berty vitamin

→ ~~कोले~~ → vit E ↓

→ Anti-oxidant

→ Digestion, Respiration → Consume

Free molecule (O) + Body cell → Damage

Dead cells:

↳ Heart → Heart Attack

Eye: cataract

Reproduction → vit E ↓ Gonads ↓

memory cells ↑

↓
~~Infertility~~
Infertility

Sources

- Dry fruits
- ~~oil~~
- Vegetable seeds oil
- cod liver oil
- Mustard oil

Vitamin - K

Phylloquinone / Naphthoquinone

↓
Discovered

↓
Dan & Norris

- Blood clotting
- Prothrombin vit K → Thrombin
- fibronogen → vit K (active)

→ vit - K → Large Intestine → E. coli
(Bacteria)

Source

पालक

- मूली

- अंडे

- chicken liver

- पत्ता गोभी, कूल गोभी

Water Soluble

Vitamin - C

- Ascorbic acid

- धातु ग्रहण में

- Immunity

- dead cell

- cancer $\rightarrow$ Vitamin C $\downarrow$

Sources

① आंवला

② नींबू

③ संतरा

④ शहदे फल

⑤ हरे आम

⑥ अमरुट

⑦ खिमला मिर्च

$\rightarrow$ धूप $\rightarrow$ Heat से Vitamin C $\downarrow$

$\rightarrow$ Iron (Fe^{2+}) $\rightarrow$ absorb

$\rightarrow$ Vitamin C $\downarrow$ $\rightarrow$ Milk में नहीं होता

$\rightarrow$ Egg में नहीं होता

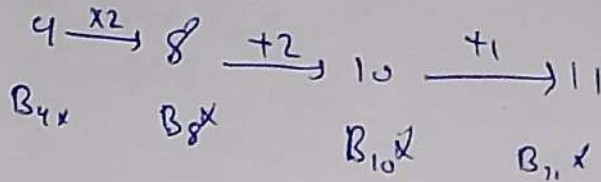
$\rightarrow$ Vit C $\downarrow$ $\rightarrow$ कैंसर $\rightarrow$ Anti oxidant

$\rightarrow$ Scurvy $\rightarrow$ मसूड़ों से रक्त आना शुरू हो जाता है

Vitamin B

Complex vitamin.

1 → 12



1 → 12 - vit B (types)

- $B_1 \rightarrow$ Thiamin थिअमिन
- $B_2 \rightarrow$ Riboflavin राइबोफ्लेविन
- $B_3 \rightarrow$ Niacin ~~नायसिन~~ ~~नायसिन~~

~~$B_4 \rightarrow$~~

$B_5 \rightarrow$ Panthothenic acid पैंथोथेनिक एसिड

$B_6 \rightarrow$ Pyridoxine पायरीडोक्सिन

$B_7 \rightarrow$ Biotin बायोटिन

~~$B_8 \rightarrow$~~

$B_9 \rightarrow$ folic acid / folate फोलिक एसिड / फोलेट

$B_{12} \rightarrow$ Cyanocobalamin सायनोकोबालामिन

Learning technique

थिअमिन, राइबोफ्लेविन, नायसिन, पैंथोथेनिक एसिड, पायरीडोक्सिन, बायोटिन, फोलिक एसिड, सायनोकोबालामिन

Brain Δ ham?n
ls $B_6; B_9, B_{12}$

$$\hookrightarrow \mathbb{B}_4; \mathbb{B}_9, \mathbb{B}_{12}$$

ଜାଣିବା ପାଇଁ : Embryo

↓
B₉ → folic acid

— Spinal chord

- C.V.N.S.

Vitamin - D₁

→ Thiamine

Veri-Veri → (analysis)

Day

De (

→ Milk, Eggs, Seeds, cashew, Rice, Wheat

→ B_1 → muscle movement

Vitamin - B₁₂

→ Riboflavin

→ Chelosis → जीवा + शीत + प्रत्यय
→ Red sea

- Red spot on skin

- milk product

vit -B₃ = Nicotin, Nikotinic acid, Niacin, acid

Source

- खाद
- Green vegetables.

Disease:-

4D (Pellegrina)

- Dermatitis → skin infection
- Dementia → memory loss
- Diarrhoea → उल्टी, दस्त
- Death

Vitamin - B₅

- Pantothenic acid
- B₅ ↓
- White hair / Grey hair
- Hair growth
- Milk product, Egg, Amla, Green vegetable

Vitamins

Vit. B₆ :-

- Pyridoxine
- RBC formation
- Brain vitamins

Source - खाद, दाल, Green vegetable

Disease:-

- Anemia
- Kidney stone
- Depression
- cramps in muscle

Vitamin B₉ :-

→ Folic acid / Folate / malic acid

- Brain vitamin
- Apple, Coconut Water
- Pregnant → B₉

↳ - Spinal chord

→ Brain development X

→ Megaloblastic Anemia → RBC Abnormal
↓
RBC

Vitamin B₇ :-

- Biotin
- Hair growth
- Baldness
- scalp infection
- यकृत ; मांस → Sulphur

Vitamin - B₁₂ :-

→ cyanocobalamin

→ cobalt → Co^{2+}

→ Pernicious Anemia

↓

→ Immature RBC

→ nervous disorder

Source - Green vegetable (spinach)

vit K, vit D₁₂

→ Large Intestine

(B. coli)